SentinelOS v1 Arizona SPO Modernization Brief Layout

Working title: `SentinelOS v1: Governed Administrative Workflow Runtime`

Audience: Arizona State Procurement Office modernization and oversight stakeholders

Format: 10-12 page PDF systems memo

Tone: measured, institutional, evidence-led

Design Guidance

- Page size: US Letter
- Margins: 0.7 to 0.9 inches
- Typography: one clean sans-serif family, 10.5-11 pt body text, 16-20 pt section headings
- Color use: restrained; use green only for verified integrity, amber for advisory/escalation, red only for refusal
- Visual style: tables, callout boxes, simple process diagrams; no promotional hero graphics
- Footer: document title, version, page number, `Prepared for discussion`

Page 1 - Cover

Title:

SentinelOS v1 Governed Administrative Workflow Runtime Vendor Onboarding Compliance Pilot Proposal

Prepared for:

Arizona State Procurement Office

Prepared by:

[Your Name / Nunn Cloud]

Date:

[Insert Date]

Footer note:

Discussion draft. Not a deployment proposal.

Page 2 - Executive Overview

Purpose:

Frame SentinelOS as governance infrastructure, not general-purpose AI automation.

Core message:

SentinelOS v1 is a governance-first administrative workflow runtime designed to automate structured compliance checks while enforcing uncertainty containment, human review, and immutable audit traceability.

Evidence callout:

Under conservative ambiguity simulation:

- Escalation rate: 3.0 percent in 1,000-case run
- Refusal rate: 0.6 percent
- Advisory band: 6.9 percent
- Ledger integrity: 100 percent
- Queue peak: 30
- Risk curve: smooth, no cliff behavior

Page 3 - Arizona Procurement Context

Purpose:

Show that the proposal is aligned with Arizona's existing procurement operating environment.

Verified context:

- The Arizona Procurement Portal is the state's e-procurement system for suppliers and state employees.
- APP supports supplier/state procurement activity across Arizona agencies, boards, and commissions.

- Arizona's APP vision includes end-to-end automation, spend management, data quality, availability, and supplier management.
- ADOA guidance says businesses must register in APP to do business with the State of Arizona.

Positioning:

SentinelOS is not a replacement for APP. It is a governance runtime layer for rule validation, escalation, and audit replay around vendor onboarding workflows.

Page 4 - Administrative AI Risk Context

Problem framing:

Public-sector workflow automation can fail when decision logic is opaque, uncertainty thresholds are hidden, human escalation is informal, audit trails are mutable, or policy versions drift from execution behavior.

SentinelOS response:

- Deterministic rule catalog
- Configurable NVOP thresholds
- Mandatory escalation review for State 2+
- Operator rationale and policy reference enforcement
- Append-only hashed ledger
- Replayable risk calculation

Page 5 - Architecture Overview

Diagram:

```
Vendor Intake -> Rule Engine -> NVOP Containment -> Latency Governor ->
Escalation Interface -> Audit Ledger -> Workflow Continuation
```

Component notes:

- Rule Engine: 30 deterministic vendor onboarding rules
- NVOP: `Risk Index = (1 - C) * I * D * (1 - V)`
- Latency Governor: assigns standard, advisory, human review, extended review, or halted class
- Operator Interface: inserts human authority with required rationale

- Audit Ledger: append-only, hash-linked decision record

Page 6 - Rule Set Structure

Table:

Group	Rule IDs	Purpose
Structural Integrity	R-001 to R-010	Required data, document, timestamp, and format checks
Policy Threshold	R-011 to R-020	Contract, insurance, certification, sanction, jurisdiction, data classification
Ambiguity & Escalation	R-021 to R-030	Overlap, proximity, conflict, metadata anomaly, partial match, sensitivity flags

Rule metadata:

Every rule includes ID, category, deterministic condition, severity, optional NVOP impact modifier, and log trigger.

Page 7 - Containment Calibration

Final locked thresholds:

State	Range	Meaning
State 0	0.0 to 0.5	Pass
State 1	0.5 to 1.3	Advisory
State 2	1.3 to 3.0	Escalation
State 3	Greater than 3.0	Refusal

Narrative:

The State 2 boundary was calibrated from 1.2 to 1.3 after stress testing showed the 1.2 posture was slightly elevated for modernization-unit presentation. The 1.3 posture preserved conservative oversight while stabilizing around a 3 percent escalation profile.

Page 8 - Simulation Evidence

1,000-case conservative ambiguity run:

Metric	Result
State 0	901
State 1	69
State 2	24
State 3	6
Escalation rate	3.0 percent
Refusal rate	0.6 percent
Queue peak	30
Ledger integrity	100 percent

1,500-case stability run:

Metric	Result
State 0	1340
State 1	113
State 2	39
State 3	8
Escalation rate	3.13 percent
Refusal rate	0.53 percent
Queue peak	47
Ledger integrity	100 percent

Chart guidance:

Include a compact bar chart for NVOP state distribution and a simple histogram summary showing concentration in low-risk bands with a thin escalation tail.

Page 9 - Operator Escalation Demonstration

Demo path:

1. Load escalated vendor case queue.
2. Open State 2 or State 3 case.
3. Show failed rules and ambiguity flags.
4. Replay NVOP formula inputs.
5. Submit decision with rationale, policy reference code, and risk acknowledgment.
6. Show decision hash and ledger hash.

Guardrails:

- No empty rationale
- No missing policy reference
- No missing risk acknowledgment
- No silent override
- No duplicate decision
- No rationale mutation after submission

Page 10 - Pilot Scope Proposal

Scope:

Vendor onboarding compliance workflow.

Duration:

90 days.

Constraints:

- No autonomous approvals
- Mandatory human review for State 2+
- Full audit visibility for modernization stakeholders
- Threshold changes only through documented governance approval
- No model retraining during pilot without approval

Deliverables:

- Escalation distribution report

- Latency distribution report
- Operator throughput report
- Incident classification summary
- Governance refinement recommendations

Page 11 - Data Handling And Governance Controls

Hosting posture:

SentinelOS can be deployed in a state-approved cloud environment with role-based access controls and data residency constraints defined during pilot scoping.

Controls:

- Centralized threshold authority
- Version-controlled rule sets
- Append-only decision ledger
- Operator documentation requirement
- Audit export readiness
- Policy replay
- Drift monitoring

Procurement positioning:

SentinelOS evaluates workflow governance and escalation discipline. It does not replace APP, supplier registration, procurement authority, or existing statutory procurement controls.

Page 12 - Meeting Objective And Next Steps

Meeting objective:

Signal mapping, not procurement approval.

Questions for Arizona stakeholders:

- Where does vendor onboarding ambiguity create friction today?
- Which escalation thresholds are politically or operationally sensitive?
- What documentation is non-negotiable for audit review?
- Where is human review mandatory today?
- What audit export format would be required?

Proposed next steps:

1. 60-90 minute technical briefing.
2. Vendor onboarding workflow mapping session.
3. Pilot governance boundary definition.
4. Controlled integration plan.
5. 90-day evaluation.

Closing statement:

SentinelOS is proposed as governed workflow containment infrastructure for administrative modernization, not as a replacement procurement system.

Source Notes

- Arizona State Procurement Office, Arizona Procurement Portal: https://spo.az.gov/arizona-procurement-portal
- Arizona State Procurement Office, Suppliers page: https://spo.az.gov/suppliers
- Arizona Department of Administration, Register To Be A State Vendor: https://doa.az.gov/about/services/register-be-state-vendor
- Arizona Procurement Portal login page: https://app.az.gov/